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Introduction to: Eye Tracking

Eye tracking measures viewers' eye movements to understand the distribution of visual attention. It can answer a variety of questions, including, but not limited to the following:

Which elements attract immediate attention?

Do some elements attract more attention than others?

Do elements get ignored or overlooked?

In what order are elements noticed?

How does one stimulus compare to other stimuli?

Do sub groups attend differently to the stimuli?

Background

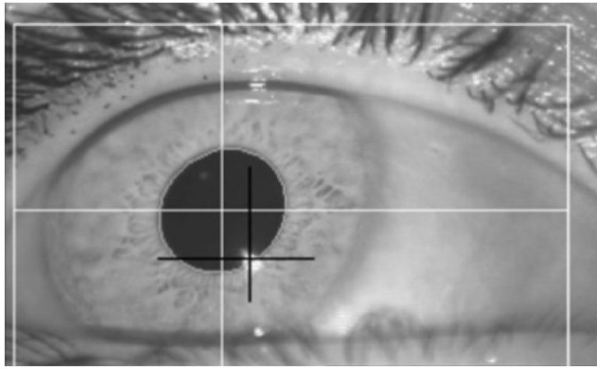
- Two main types of eye tracking hardware are screen based eye trackers and eye tracking glasses. Eye tracking can also be utilized in virtual environments using VR-ET headsets that are outfitted with eye tracking hardware.
- - Screen-based eye trackers record eye movements at a distance. The eye tracker is mounted below or placed close to a computer screen, and the respondent is seated in front of the eye tracker.
 - Eye tracking glasses have eye tracking technology mounted onto lightweight eyeglass frames that allow the respondent to walk freely, while eye activity is recorded from a close range.
- While screen-based eye trackers are recommended for observations of any screen-based stimulus, such as pictures, videos, websites and offline stimuli, glasses and VR-ET headsets are used in real-life or virtual environments, respectively, for use cases such as human factors and product testing.
 - Screen-based eye trackers can also be utilized for testing on mobile devices. A mobile stand holds the eye tracker and mobile device, and extra cameras can be added to monitor the respondent as well.



Screen-Based Eye Trackers (left) and Eye Tracking Glasses (right)

How it Works

- The underlying concept behind ET is Pupil Center Corneal Reflection, or PCCR. This process involves the camera tracking the pupil center and where the light reflects from the cornea, informing the eye tracker about the movement and direction of the eye.
- The eye is illuminated using infrared light, which is not perceivable by the human eye. With some of the light entering the pupil and the rest of the light being reflected by the eyeball, one obtains a clear contrast between the pupil and iris. This way, a live image processing algorithm can detect the pupil as well as the light reflections on the cornea. When the gaze direction is changed, the relative position of the pupil and corneal reflexes change, which enables the eye tracker to compute where the respondent is looking at.

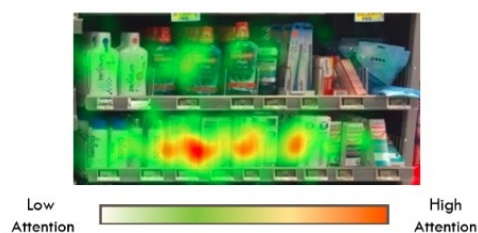


Quantification

- Eye tracking is quantified in various different ways. For our purposes, we will focus on fixation and gaze points.
 - Eye movements between fixations are known as saccades. While reading a book, for example, eyes do not move completely smoothly across a line, but instead jump and pause thereby generating a vast number of discrete sequences.
 - Gaze points constitute the most basic unit of measure. One gaze point equals one raw sample captured by the eye tracker. If the eye tracker measures 60 times a second, then each gaze point represents a sixtieth of a second, or 16.67 milliseconds.
 - If a series of gaze points happen to be close in time and range, the resulting gaze cluster denotes a fixation, a period in which our eyes are locked toward a specific object. Typically, the fixation duration is 100-300 milliseconds.
 - A fixation, therefore, may indicate an instance in which a component of a stimulus captures our visual attention, and is thus a useful measure to use to quantify eye tracking data and derive meaningful insights from that data.

Data Visualizations

- Heatmaps are static or dynamic aggregations of gaze points or fixations revealing the distribution of visual attention. In iMotions, researchers can choose gaze points or fixations depending on their preference.
- Visual attention is aggregated across the entire audience, and displayed as “heat” overlaid on a stimulus. This heat can help identify what areas of the content draw the most visual attention, and what areas or elements may go missed entirely.
 - While red areas suggest a high number of gaze points, and therefore an increased level of interest, yellow and green areas show fewer gaze points, indicating a less engaged visual system.



- Spotlight maps are generated in the same way as a heatmap with a different color scheme. In these maps, pictures are overlaid with a black layer, and areas that receive visual attention are more “lit up” the more attention that was directed in that area (like a spotlight). These maps provide a different way of visualizing attention across your stimulus.
- These kinds of representations are useful at providing an overview of how visual attention is distributed across a particular scene or stimulus.
 - For a deeper look at how heatmaps are generated, please review our help center article [here](#).
 - For this reason, it is important to utilize other available tools, such as areas of interest, to delve deeper into quantitative aspects of your tested stimulus.
 - While heatmaps and spotlight maps can be incredibly useful for visual representation and determining which aspects of the stimulus were most relevant (or overlooked) by the entire sample, it is important to understand that these are purely qualitative visualizations. To interpret these maps appropriately, it is useful to understand how they are calculated.
- Areas of Interest (AOIs) are user-defined subregions of a displayed stimulus. AOIs are an essential tool to incorporate into your analysis for quantification and a more objective approach to interpreting eye tracking data¹. For example, AOIs may be used to evaluate the performance of two or more specific areas in the same video, picture, website or program interface. AOIs can be utilized to compare groups of participants, conditions, or different features within the same scene using time spent, respondent count, fixation sequences, or time to first fixation metrics.
 - For example, these metrics can reveal how many respondents fixated in that area of interest, how long it took them to fixate in that area, and how long they spent looking. These metrics can indicate if a stimulus is capturing visual attention in the intended way, or if the stimulus can be optimized to better appeal to natural visual patterns.



- For example, perhaps you have created a website with a particular blurb that includes your value proposition. After running a biometric eye tracking test on the UI of that website, AOI's may reveal that only ¼ of respondents ever made it to that section, and when they did, they only spent an average of 0.5 seconds there - not nearly enough time to read and fully process the value prop. These metrics indicate that the layout of the website can be optimized to draw attention to the value prop - perhaps by switching placement with another area that drew attention from more viewers, and for longer.
- Areas of interest can also provide useful information on the order in which elements of interest were navigated by the sample.
 - For example, you may draw an AOI around the main products on the shelf you have tested. These AOIs can indicate, within that stimulus, a ranking of how long it takes respondents on average to look at those objects for the first time.



Image created manually by ranking the average TTFF of each area of interest from first seen to last seen.

- Scanpaths superimpose individual gaze paths on a stimulus to tell us how respondents explored the material. These can be viewed in various ways:

Dynamic recordings

- For example, perhaps a participant looks back and forth between certain objects or had to look back to a certain picture while trying to understand text. This observation may prompt the researcher to probe further to understand if it may have been difficult to understand the difference between those objects or to comprehend the text they were reading.
- This "look back" behavior may also be quantified as a refixation in that area of interest.
- One way to view a scanpath is by viewing the dynamic recording of the gaze of the respondent as they explored the material. This technique can be utilized for stimuli like static images, videos, or real-life and virtual environments.
- Often, individual scanpaths are utilized in biometrically informed interviews, in which moderators observe the experience of an individual participant, and utilize the replay of that individual's data to guide questions in the qualitative interview to follow.

Static representations

- Another way to view scanpath data is by gaze mapping, or mapping the gaze data on to a static representation of the stimulus. This technique would be utilized, for example, in mobile eye tracking glasses studies in which researchers would like to map the natural visual exploration of a shelf onto a static image representation of that shelf in order to analyze that data. This can be done for individual or group data.

Scanning behaviors can provide information on navigation patterns and reveal how a material may be directing an audience's gaze. In UX and UI testing, for example, researchers will often take a task-based approach in which participants are instructed to complete a certain task, such as purchasing a product or locating a specific set of information. Scanpaths can show if consumers are taking the "optimal" path through the website to complete the task, or if there are specific pain points or sources of confusion that can be optimized.

What ET Does Not Measure

- While eye tracking provides a wealth of information on where and when an individual looked, lengths of fixations, and the order in which elements are fixated upon, ET does not reveal *why* an individual looked at a certain element, or *how they felt* while navigating a certain visual scene. For this reason, ET is best utilized when combined with other biosensor methods to reveal physiological arousal, behavioral motivation, expressed emotion, or self-reported thoughts and feelings in relation to a stimulus or experience.

Academic Applications

Historically, eye tracking has been used across a range of different research fields, and for various applications within the commercial realm, too. In psychology, understanding when and how people look is essential for understanding how attention is distributed. For example, ET is used in the Implicit Association Test, Stroop Test, and Iowa Gambling Task, as well as in other psychological explorations to gain deeper insights into cognitive processes underlying attention, learning, and memory. In the medical field, ET can be used to diagnose neurological disorders such as ADHD, Autism Spectrum Disorder, OCD, Schizophrenia, Parkinson's, and Alzheimer's disease, and continues to be utilized to further research in these fields. Additionally, ET can be used to detect states of drowsiness or support other fields of medical use, quality assurance or monitoring, and these fields continue to expand and grow into new and exciting research areas.

Commercial Applications

Over the last few decades, eye tracking has become increasingly important for market research. Many leading brands use this tool to evaluate their products, designs, and advertising campaigns. Commercial researchers utilize both mobile eye tracking glasses and screen-based solutions to investigate the shopping behavior of their customers with the goal of optimizing the overall customer experience.

In neuromarketing, analyzing what people attend to or ignore and how visual attention is distributed can be critical in determining optimal package designs, store layouts, and point-of-sale displays. Eye tracking for usability and user experience is also an emerging field using these methodologies. One prominent use case is website testing. Commercial researchers employ eye tracking to investigate gaze patterns on old and new website designs to measure, for example, attention to real estate, communication, and calls to action (CTA). Similar applications can be applied to mobile apps on tablets and smartphones, too.

Eye tracking is also used in packaging research, specifically pertaining to the design of packages to understand what grabs attention, which elements or information may go missed, and customers' overall preferences. In this way, researchers aim to ensure that the package for a product receives visual attention on the shelves, and effectively draws attention to the salient images and information on the package itself. Optimizing the way a package draws attention and the way this attention is distributed may be contributing factors in drawing consumers to purchase.

Eye tracking methodology has also been introduced within the human-computer interaction, virtual reality, and gaming industry space to enable game designers to better understand the game experience. These insights may allow designers to better control the experience, optimize usability, and create features that further push the boundaries of reality and improve the gaming experience of users.

In the media industry, eye tracking is utilized to measure attention distribution on media ads, trailers, full films, television shows, and more. While GSR is utilized to provide both overall and moment to moment insights on the emotional arousal and engagement of an audience throughout a piece of creative, eye tracking can provide a view into how visual attention is distributed in key moments.

Eye tracking can provide a wealth of information about the visual experience of consumers and users across a wide range of informational materials, products, and experiences. Consider adding eye tracking as a methodology to your research to complement traditional methods.

¹Typically, we would recommend a sample size of at least 20 respondents to capture reliable area of interest data.

For more in-depth explanations of eye tracking as a metric, please check out these posts on our blog:

[What is Eye Tracking and How Does it Work](#)

[Anatomy of an Eye Tracker \[How Eye Tracking Works\]](#)

[How to Analyze and Interpret Heatmaps](#)

[10 Most Used Eye Tracking Metrics and Terms](#)

For use cases and detailed examples of how ET is used in commercial research, please download our [sample reports](#).

